

Chapter 16 (Carey Chapter 13)

The Chemistry of Benzene and Its Derivatives

Summary of reactions in this chapter (Carey chapter 13)

All electrophilic aromatic substitution reactions occur by similar mechanisms.

TABLE 13.1 Representative Electrophilic Aromatic Substitution Reactions of Benzene

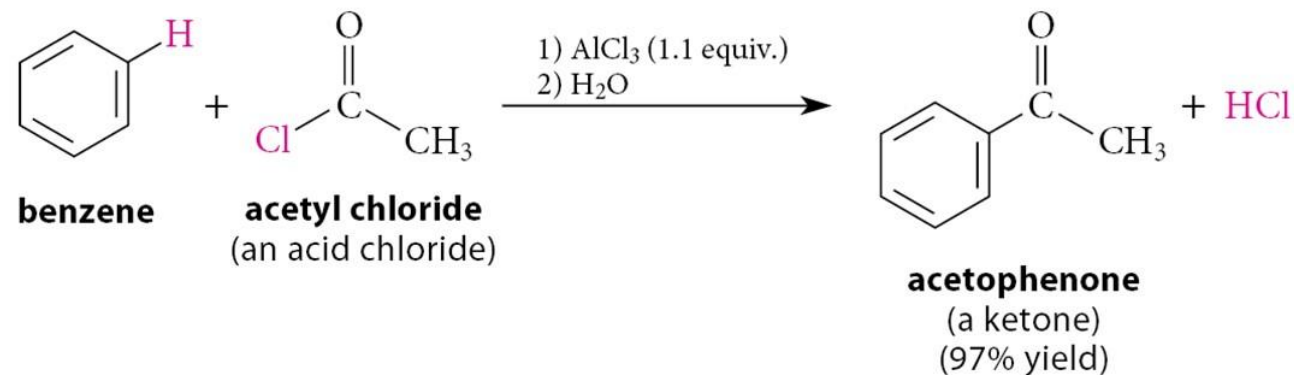
Reaction and comments	Equation
1. Nitration Warming benzene with a mixture of nitric acid and sulfuric acid gives nitrobenzene. A nitro group ($-\text{NO}_2$) replaces one of the ring hydrogens.	Benzene + Nitric acid $\xrightarrow[30-40^\circ\text{C}]{\text{H}_2\text{SO}_4}$ Nitrobenzene (95%) + Water
2. Sulfonation Treatment of benzene with hot concentrated sulfuric acid gives benzenesulfonic acid. A sulfonic acid group ($-\text{SO}_2\text{OH}$) replaces one of the ring hydrogens.	Benzene + Sulfuric acid $\xrightarrow{\text{heat}}$ Benzenesulfonic acid (100%) + Water
3. Halogenation Bromine reacts with benzene in the presence of iron(III) bromide as a catalyst to give bromobenzene. Chlorine reacts similarly in the presence of iron(III) chloride to give chlorobenzene.	Benzene + Bromine $\xrightarrow{\text{FeBr}_3}$ Bromobenzene (65–75%) + Hydrogen bromide
4. Friedel–Crafts alkylation Alkyl halides react with benzene in the presence of aluminum chloride to yield alkylbenzenes.	Benzene + <i>tert</i>-Butyl chloride $\xrightarrow[0^\circ\text{C}]{\text{AlCl}_3}$ <i>tert</i>-Butylbenzene (60%) + Hydrogen chloride
5. Friedel–Crafts acylation An analogous reaction occurs when acyl halides react with benzene in the presence of aluminum chloride. The products are aryl ketones.	Benzene + Propanoyl chloride $\xrightarrow[40^\circ\text{C}]{\text{AlCl}_3}$ 1-Phenyl-1-propanone (88%) + Hydrogen chloride

Activation by
dehydration with
acid (H^+)

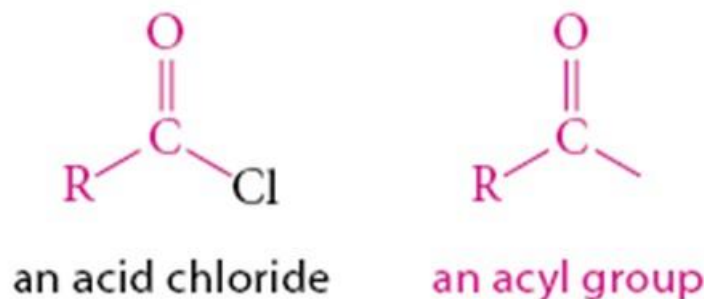
Activation by
halogen abstraction
with Lewis acid

16.4 Electrophilic Aromatic Substitution Reactions of Benzene

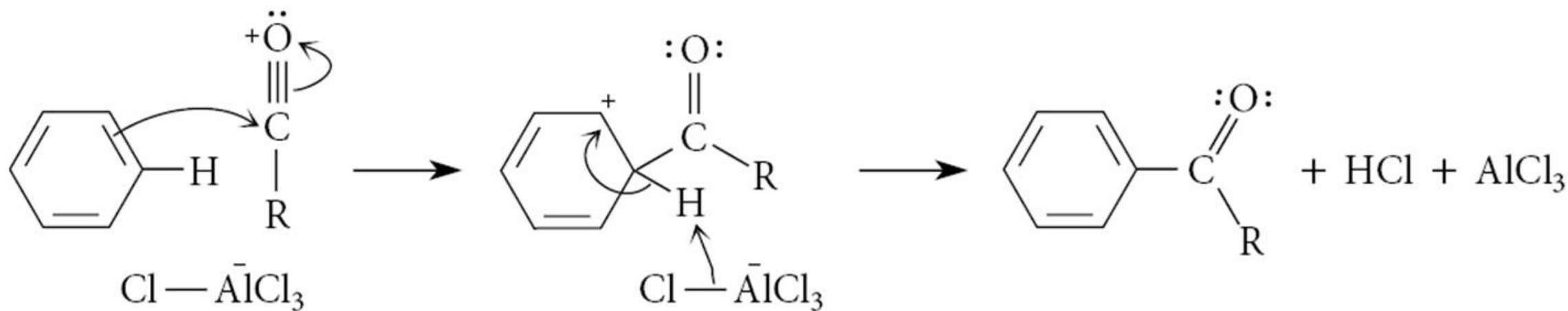
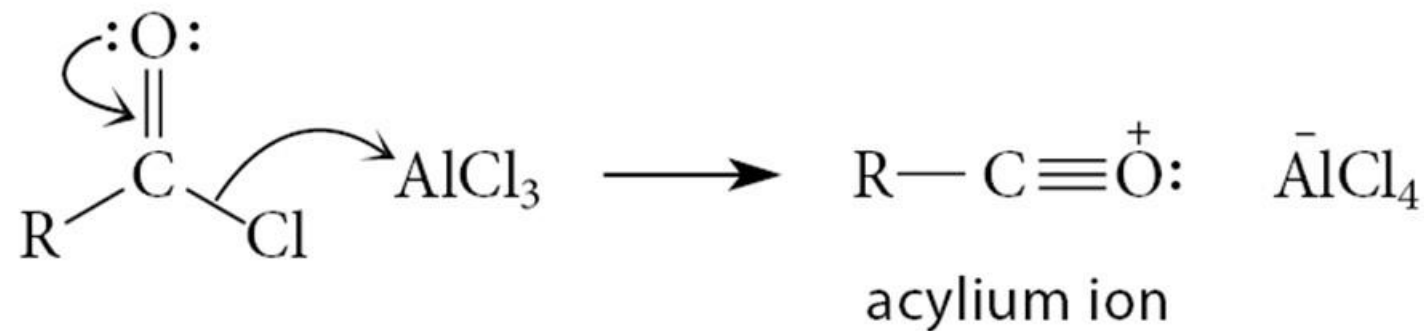
Friedel–Crafts Acylation of Benzene



- The acyl group is typically derived from an acid chloride.
(acyl consists of C=O, carbonyl group)

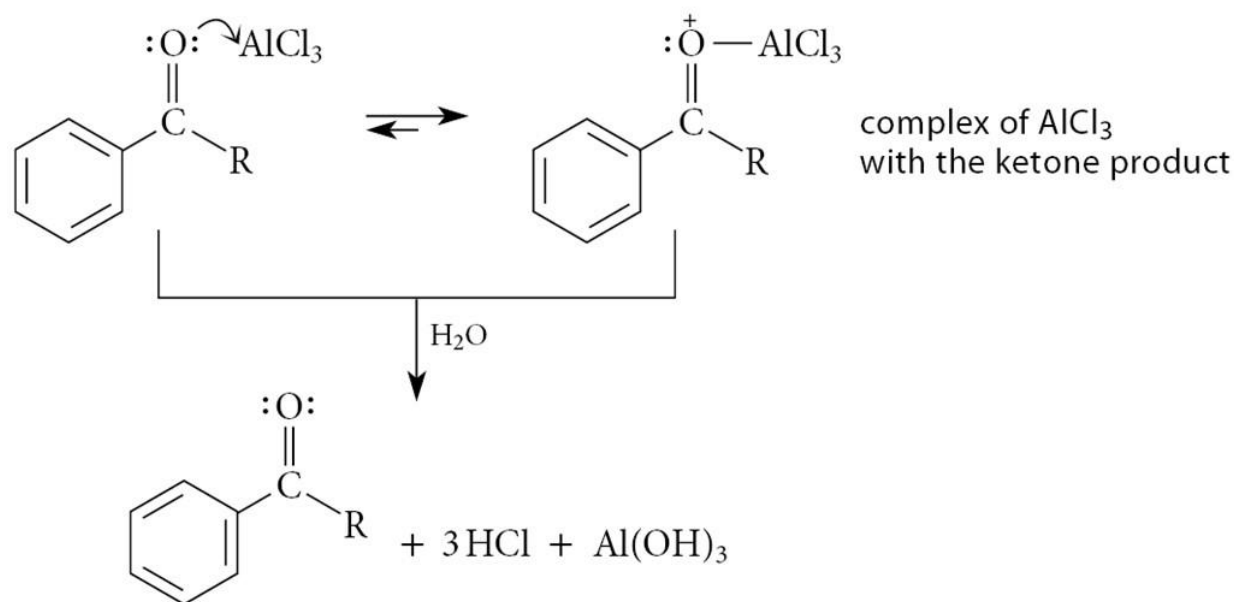


Friedel–Crafts Acylation of Benzene: Mechanism



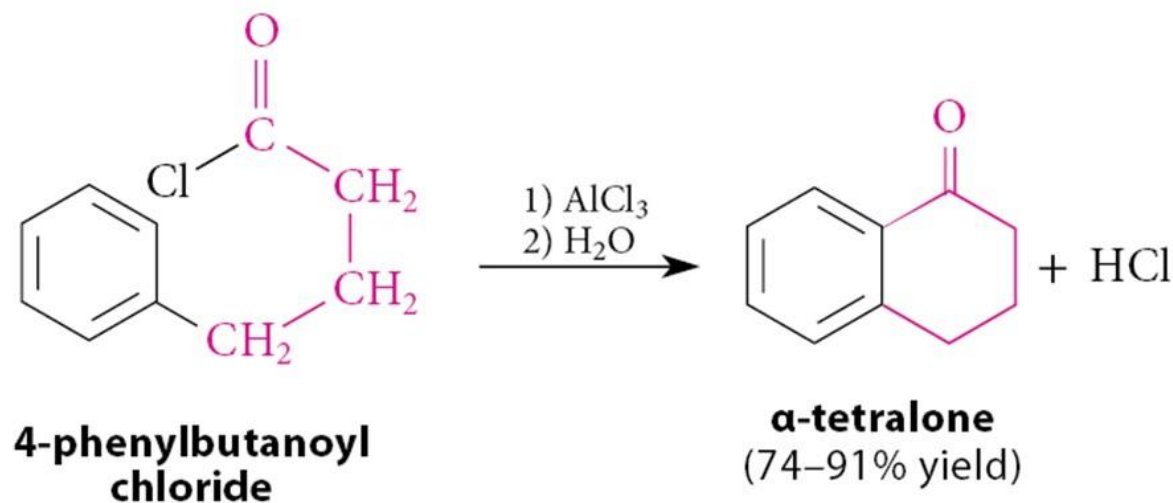
Friedel–Crafts Acylation of Benzene: AlCl_3

- The alkylation reaction uses AlCl_3 in *catalytic* amounts. In contrast, the acylation requires one *equivalent* of AlCl_3 .



Intramolecular Friedel–Crafts Reactions

- Both the alkylation and acylations can occur intramolecularly.
- The reaction occurs at an *ortho* position and is best for five- and six-membered rings.



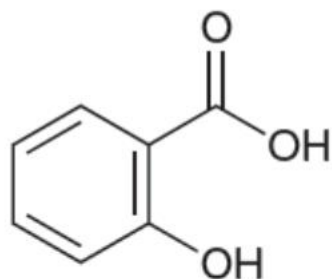
Summary of Reactions that Form C–C Bonds

- Org I {
1. Reaction of acetylenic anions with alkyl halides or alkyl sulfonates
 2. Reaction of Grignard reagents with ethylene oxide and lithium organocuprate reagents with epoxides
 3. Addition of carbenes and carbenoids to alkenes
 4. Diels–Alder reactions
 5. Friedel–Crafts reactions

Substitution positions of benzene derivatives

- The same chemical formula (isomers), but different properties.

ortho-hydroxybenzoic acid



Salicylic acid

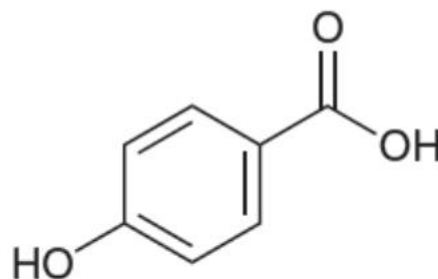
$pK_a = 3.0$

(stronger acid)

Used for exfoliation.

A precursor of aspirin

para-hydroxybenzoic acid



para-Hydroxybenzoic acid

$pK_a = 4.6$

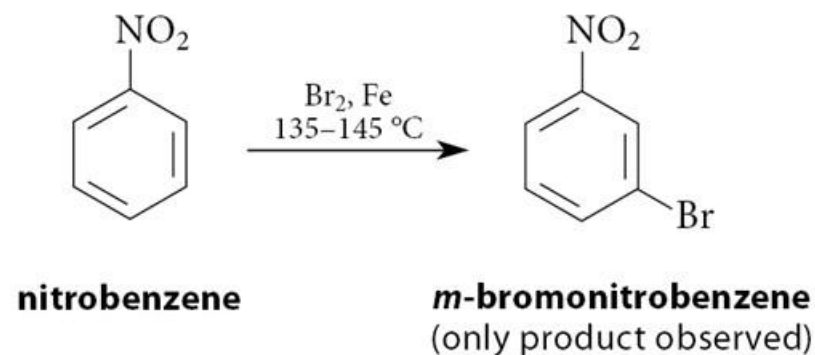
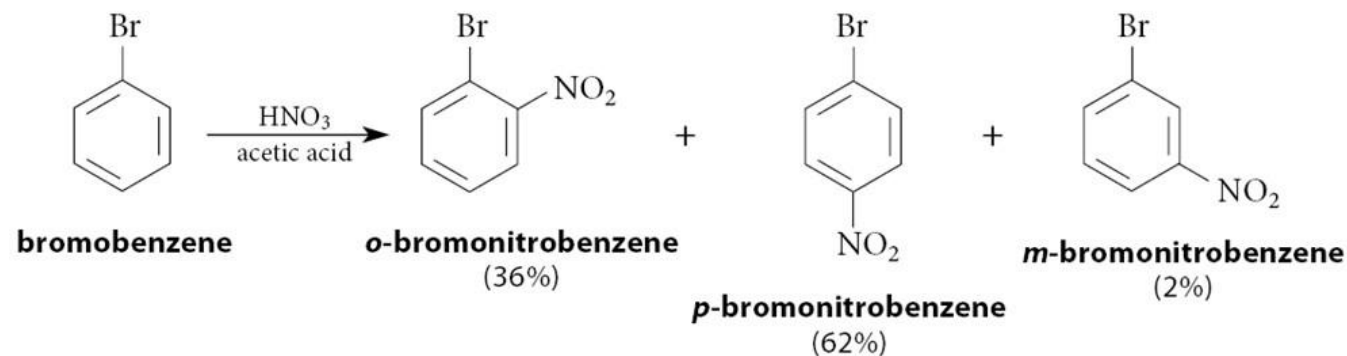
Anti-oxidant activity

Estrogenic

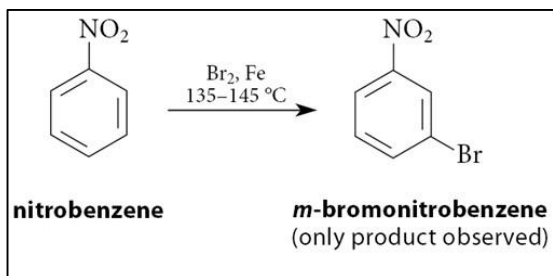
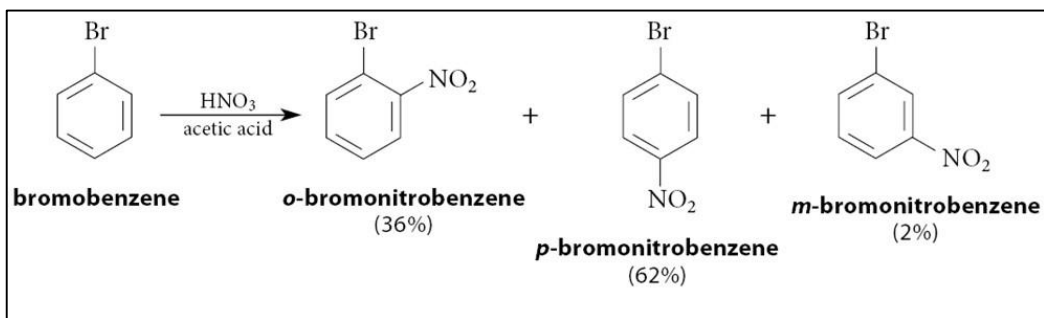
- Importance to make one isomer selectively.
- How can we control the reaction on a particular position?

Directing Effects of Substituents

- Regioselectivity is observed in EAS reactions involving *monosubstituted* benzene rings.
- Ortho*-, para-directing:
- Meta*-directing:



Directing and Activating Effects



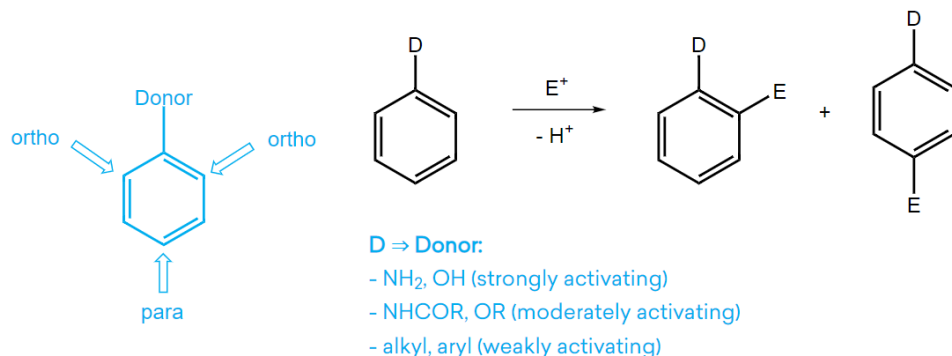
Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$\left\{ \begin{array}{l} \text{C}(=\text{O})\text{OH} \\ \text{C}(=\text{O})\text{NH}_2 \\ \text{C}(=\text{O})\text{OR} \end{array} \right\}$	carboxy, carboxamido, carboalkoxy	meta directors	deactivating substituents
$-\text{C}(=\text{O})\text{R}$	acyl	meta directors	deactivating substituents
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

16.5 Electrophilic Aromatic Substitution Reactions of Substituted Benzenes

Directing and Activating Effects

General
Chemistry 1 ▾General
Chemistry 2 ▾General
Chemistry 3 ▾Organic
Chemistry 1 ▾Organic
Chemistry 2 ▾

Activated Benzene:

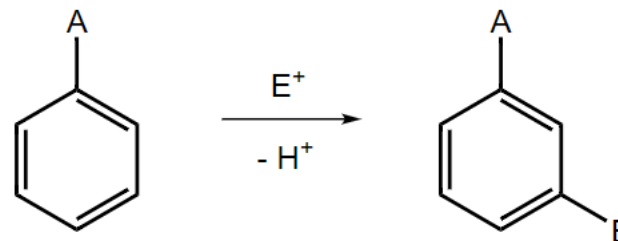
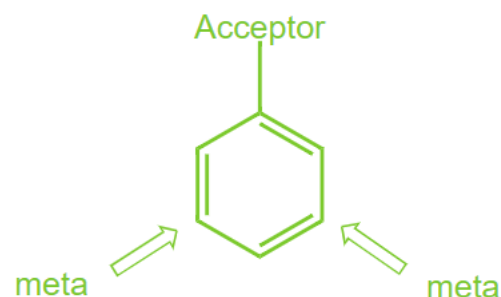


- Electron-donating groups (often atom with a lone pair directly attached to benzene): ortho and para directing.
- (exception of halogen will be learned later)

Substituent group	Name of group	Directing effect	Activating or deactivating
$-\text{NH}_2, -\text{NR}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\text{OH}$	hydroxy		
$-\text{OR}$	alkoxy		
$-\text{NH}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl		
$-\text{O}-\text{C}(=\text{O})\text{R}$	acyloxy	ortho, para directors	deactivating substituents
$-\text{C}_6\text{H}_5$	phenyl		
$-\text{F}, -\text{Cl}, -\text{Br}, -\text{I}$	halogens		

Directing and Activating Effects

Deactivated Benzene:



A ⇒ Acceptor:

- N^+R_3 , NO_2 , CF_3 , CN , SO_3H (strongly deactivating)
- COH , COR , COOH , COOR , CONH_2 (moderately deactivating)

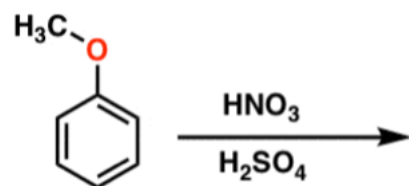
- Electron-withdrawing groups (acceptor): meta directing.
- Often, atom with multiple bond attached to benzene (no available lone pairs)

$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \\ \\ \text{R} \end{array}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

meta directors

deactivating substituents

Reaction of methoxybenzene (OMe)?

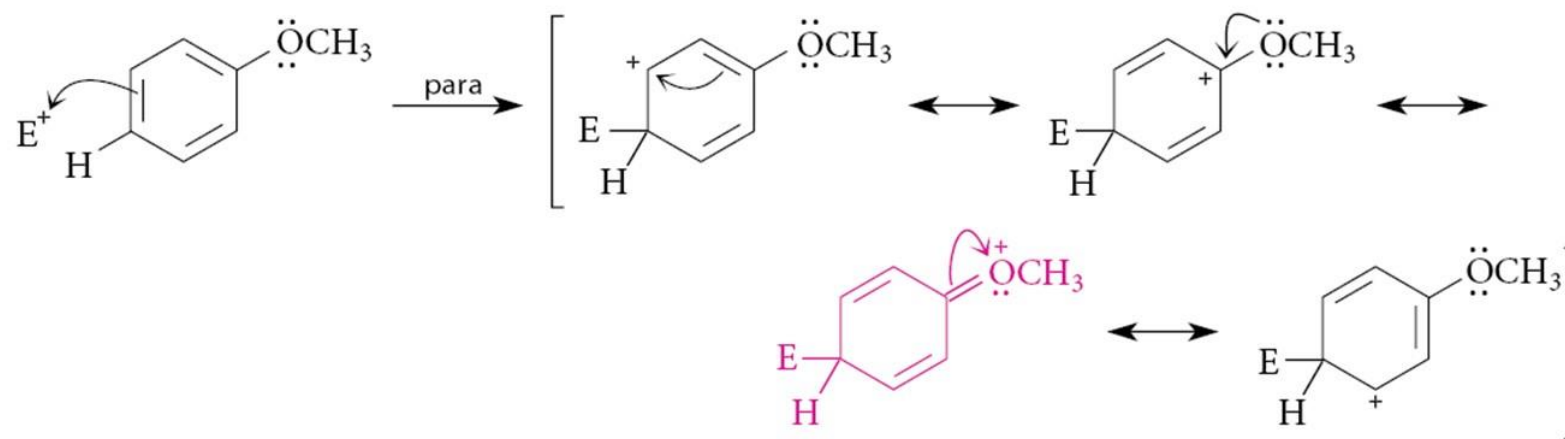


<https://www.masterorganicchemistry.com/2018/01/29/ortho-para-and-meta-directors-in-electrophilic-aromatic-substitution/>

Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$\left\{ \begin{array}{l} -\text{C}(=\text{O})\text{OH} \\ -\text{C}(=\text{O})\text{NH}_2 \\ -\text{C}(=\text{O})\text{OR} \end{array} \right\}$	carboxy, carboxamido, carboalkoxy	meta directors	deactivating substituents
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

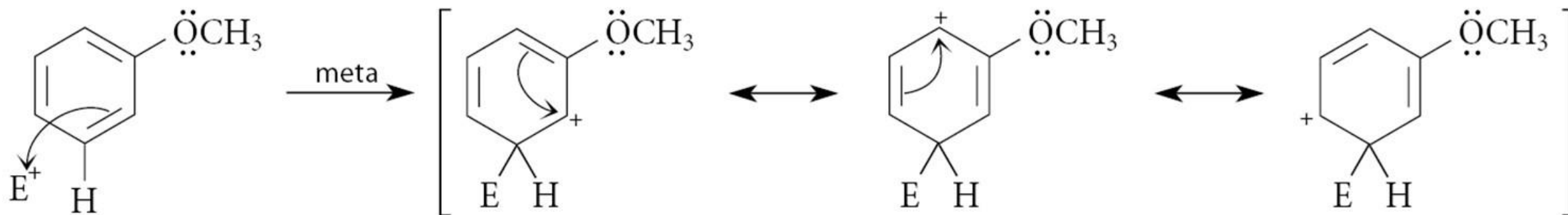
Ortho-, Para-Directing Groups (1 of 2)

- Alkyl groups **or** groups that have unshared electron pairs on atoms directly attached to the benzene ring.
- Charge of intermediate is delocalized when reaction occurs at *ortho* or *para* positions



Ortho-, Para-Directing Groups (2 of 2)

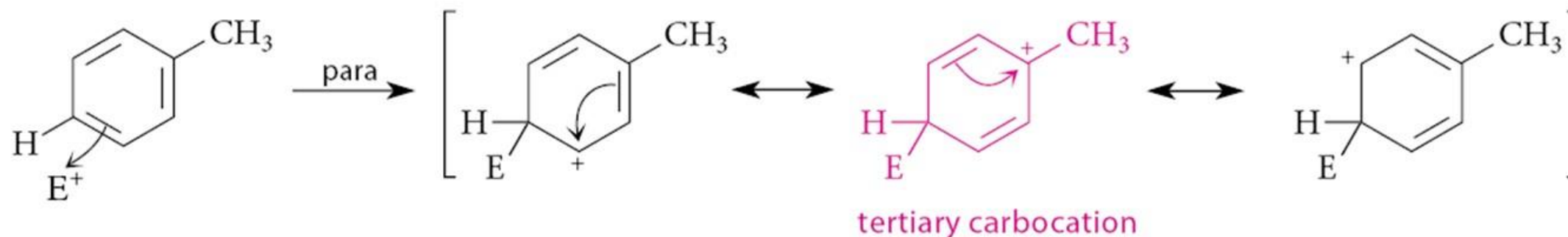
- If reaction occurs at the *meta* position, the charge cannot be delocalized onto the $-\text{OCH}_3$ group.



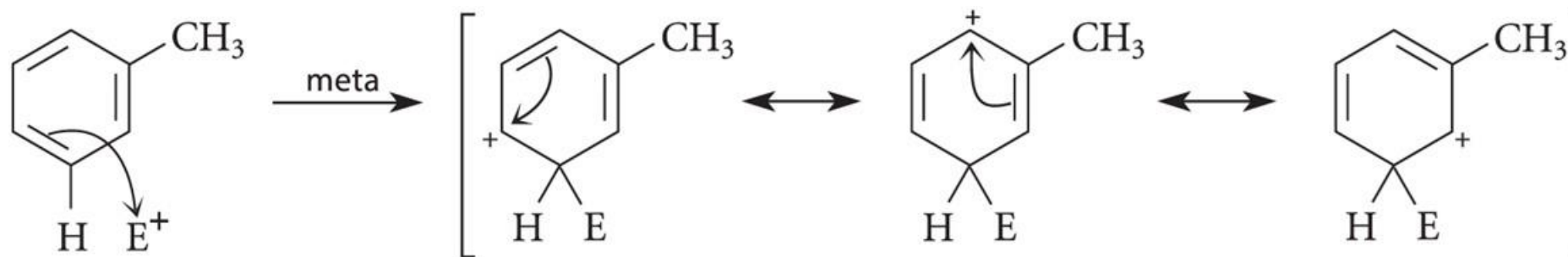
- Hence, the intermediate and transition state is less stable for reaction at the *meta* position

Alkyl Groups Also Stabilize the Carbocation Intermediate (hyperconjugation)

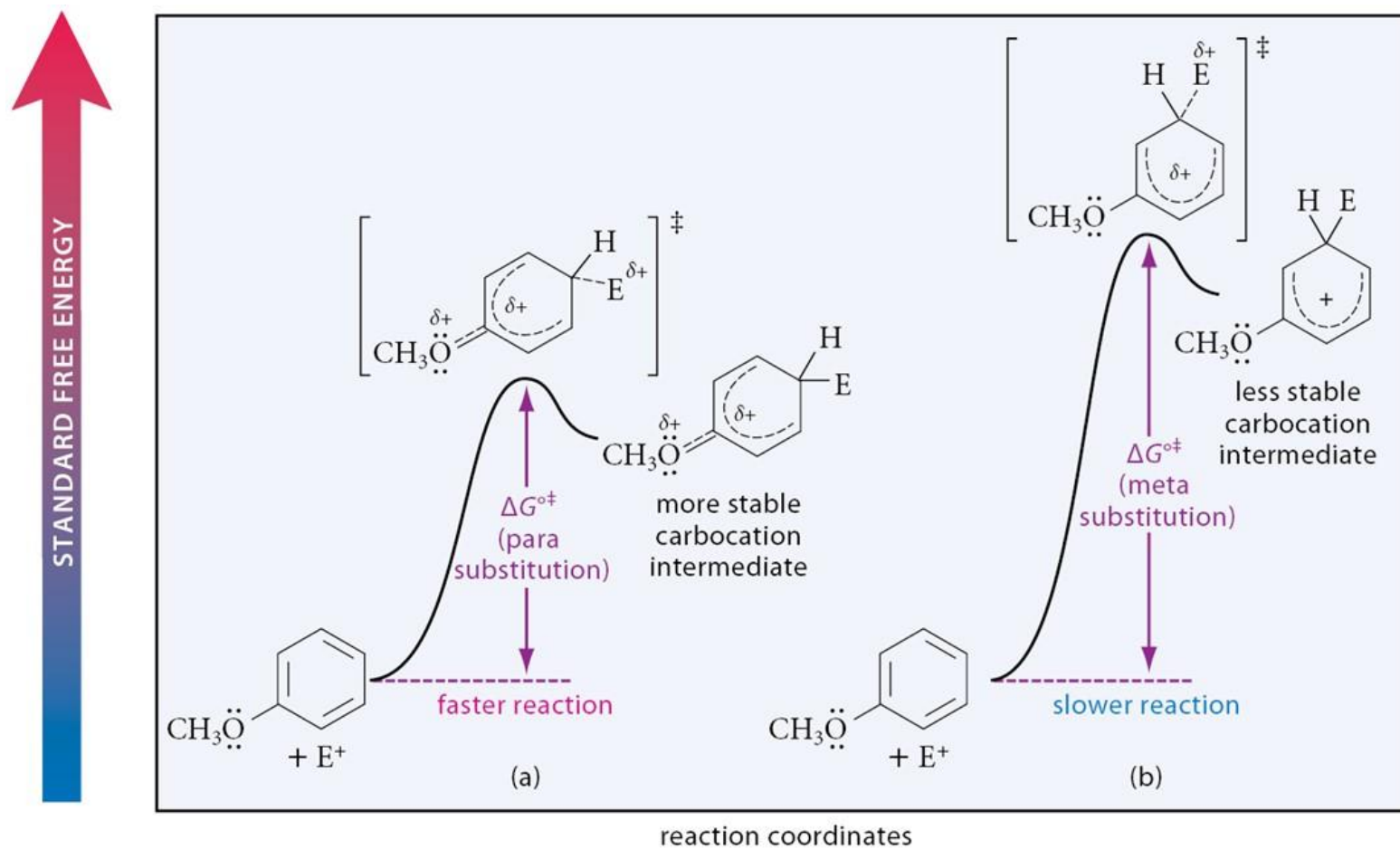
- Reaction at *para* position:



- Reaction at *meta* position:



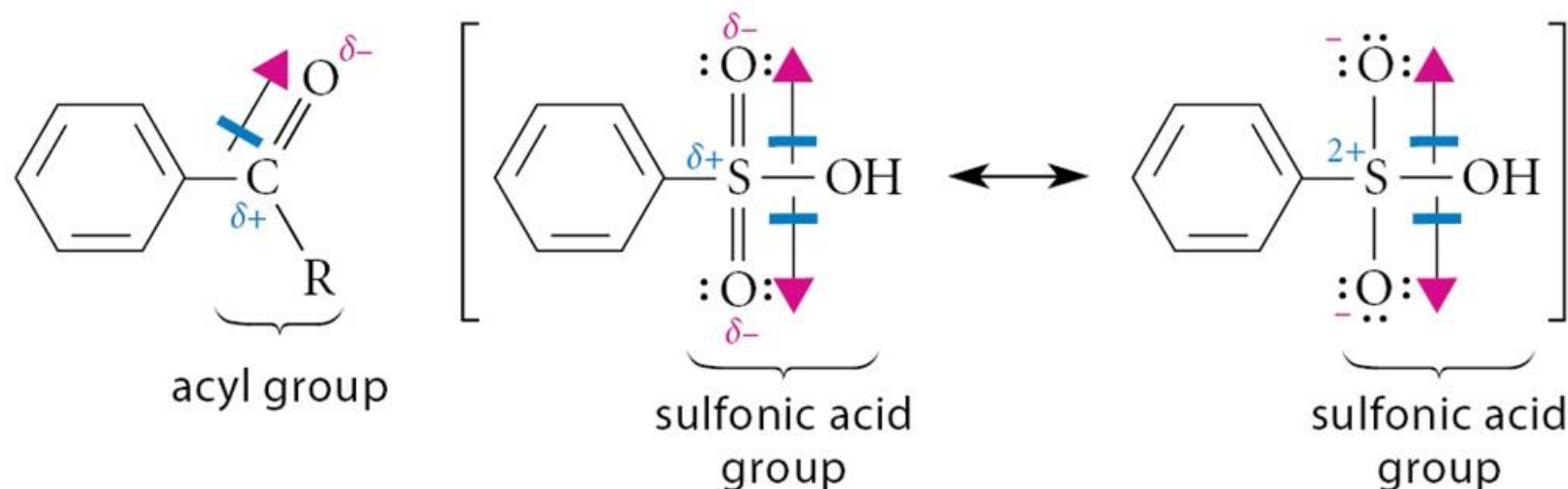
Directing Effect of the Methoxy Group



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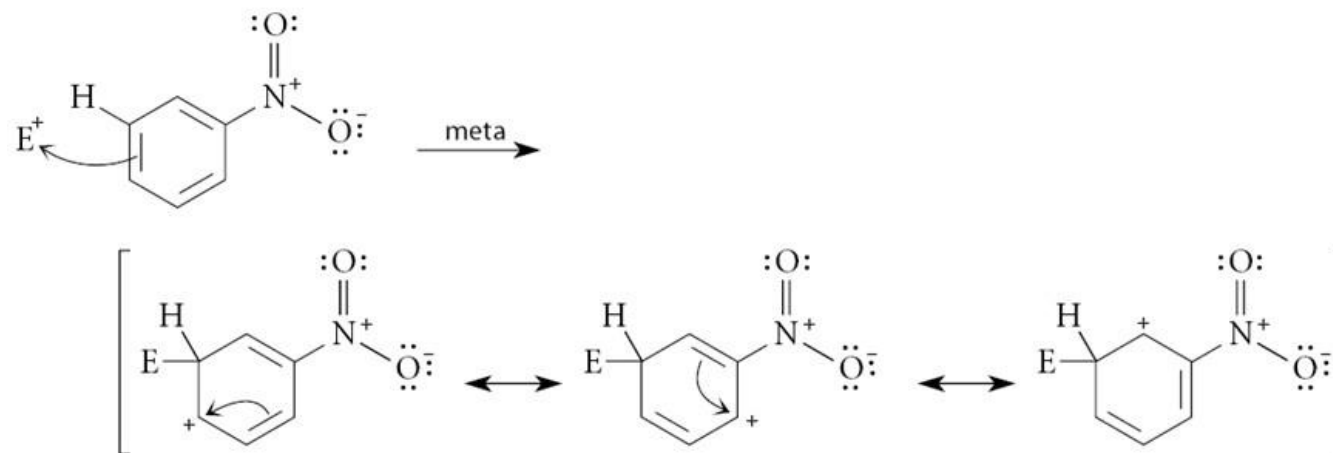
Meta-Directing Groups

- Are polar groups without an unshared electron pair on an atom adjacent to the benzene ring.
- Have positive or partial positive charge on the atom adjacent to the benzene ring.
- Examples:

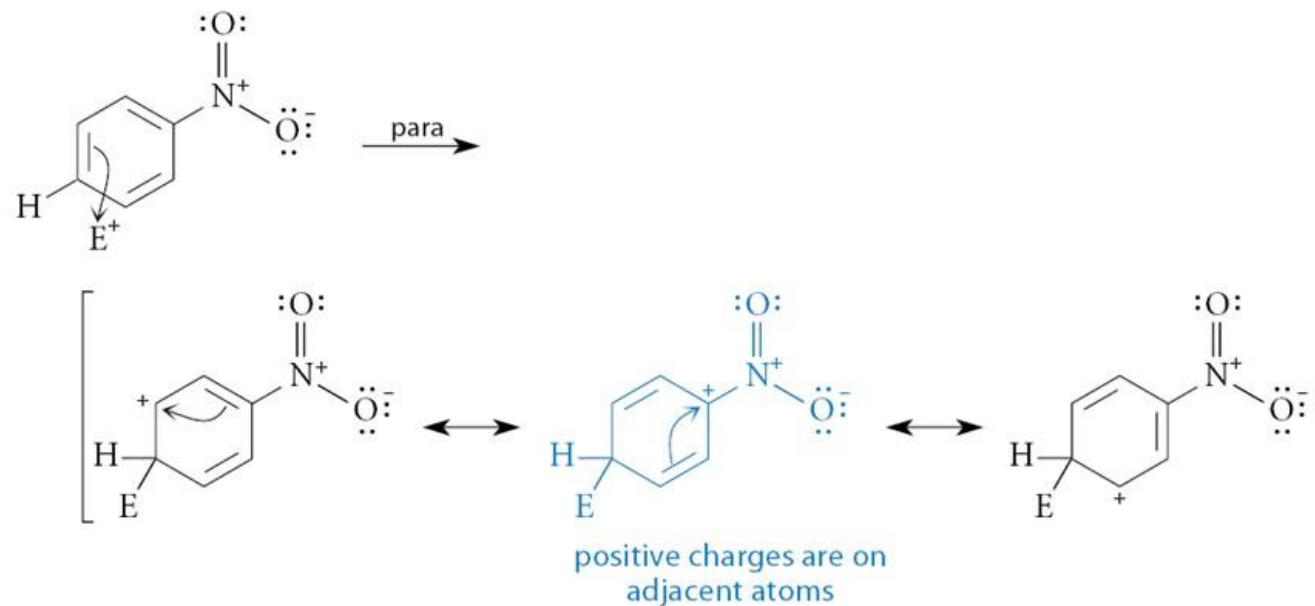


Directing Effects of Nitro Group

- *Meta:*



- *Para:*



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The *Ortho*, *Para* Ratio

- Twice as much *ortho* product as para is expected if substitution is completely random. This is rarely observed; *para* typically predominates over *ortho*.
- Sometimes this is due to spatial demands, but many cases are less easily explained.
- Fortunately, *ortho* and *para* products have different physical properties and can be separated.

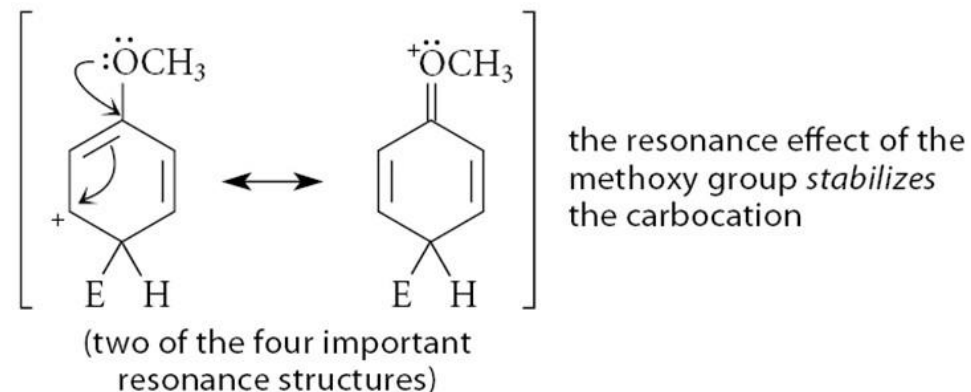
Activating and Deactivating Effects

- If a substituted benzene derivative reacts more rapidly than benzene itself, then the substituent is said to be an ***activating group***.
- If a substituted benzene derivative reacts more slowly than benzene itself, then the substituent is said to be a ***deactivating group***.
 1. All *meta*-directors are deactivating.
 2. All *ortho*-, *para*-directors (except halogens) are activating.
 3. Halogens are deactivating.

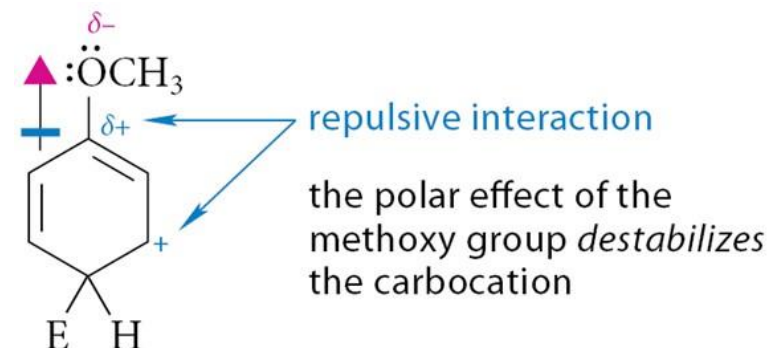
Explaining Activation– Deactivation

- Two simultaneously operating properties of substituents control the activating and deactivating effects: resonance and polarity.

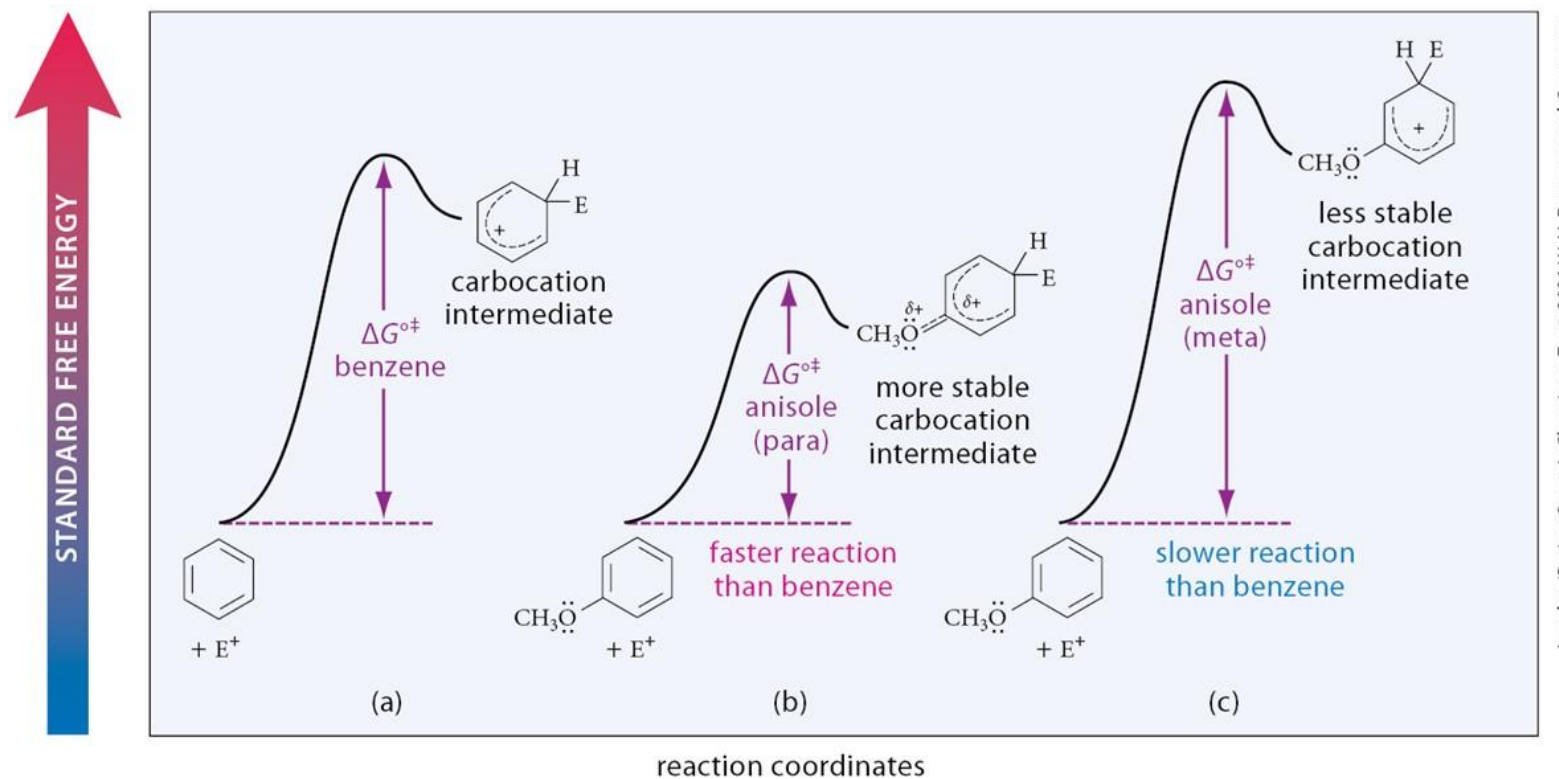
- Resonance effect:**



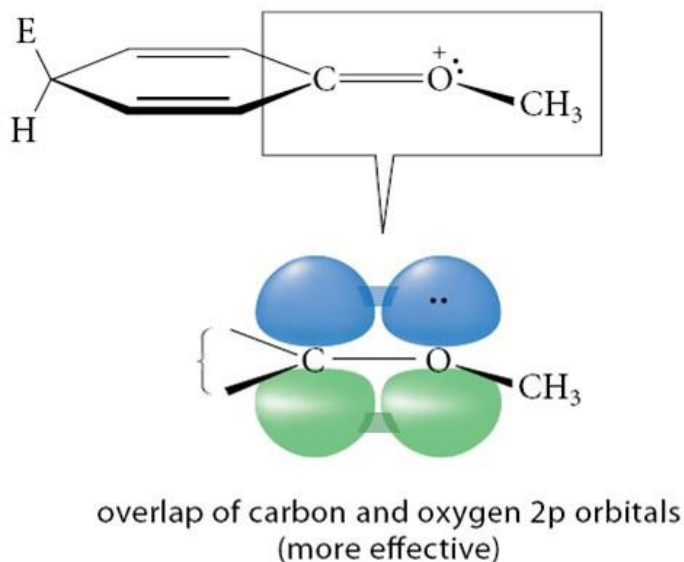
- Polar effect:**



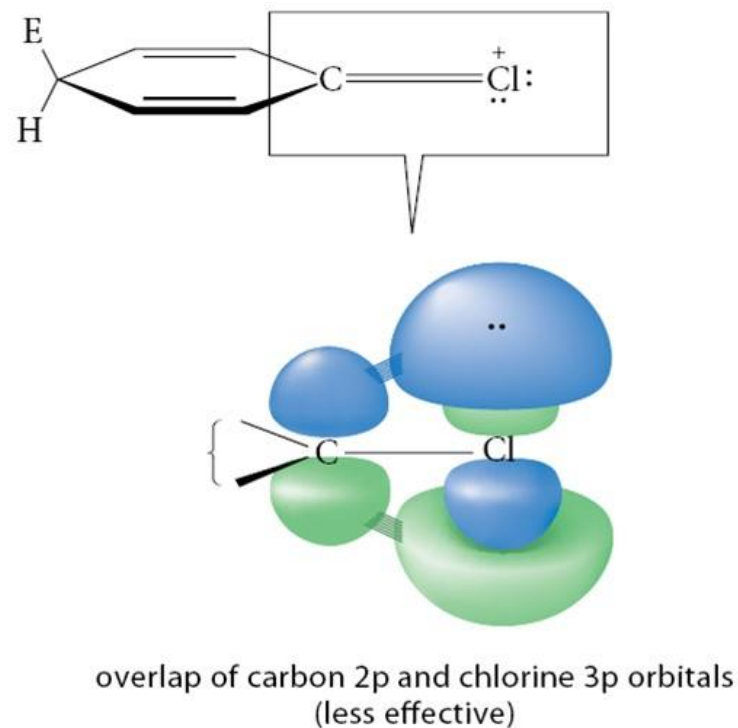
Standard Free Energy Diagram of Activating and Deactivating Effects



Orbital Overlap May Affect Degree of Resonance Effect



(a)

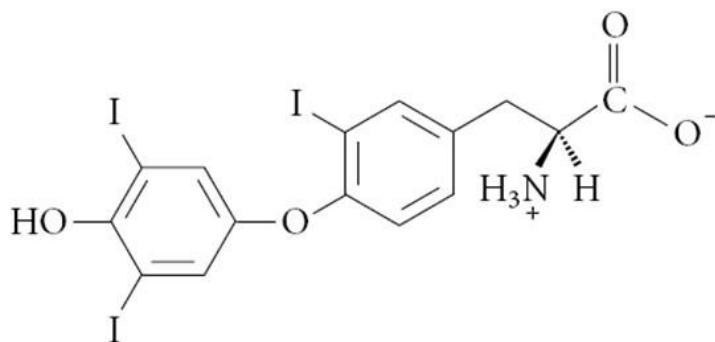


(b)

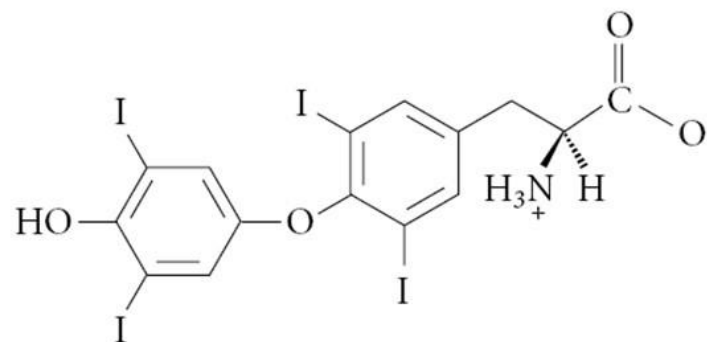
Substituent group	Name of group	Directing effect	Activating or deactivating
$-\text{NH}_2, -\text{NR}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\text{OH}$	hydroxy		
$-\text{OR}$	alkoxy		
$-\text{NH}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\text{O}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\text{F}, -\text{Cl}, -\text{Br}, -\text{I}$	halogens		
$-\text{C}(=\text{O})\text{OH}, -\text{C}(=\text{O})\text{NH}_2, -\text{C}(=\text{O})\text{OR}$	carboxy, carboxamido, carboalkoxy	meta directors	deactivating substituents
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Electrophilic Aromatic Substitution in Biology: Thyroid Hormone Synthesis

- Electrophilic aromatic substitution is involved in the biosynthesis of thyroid hormones.
- T3 and T4 thyroid hormones account for the dietary iodine requirement.



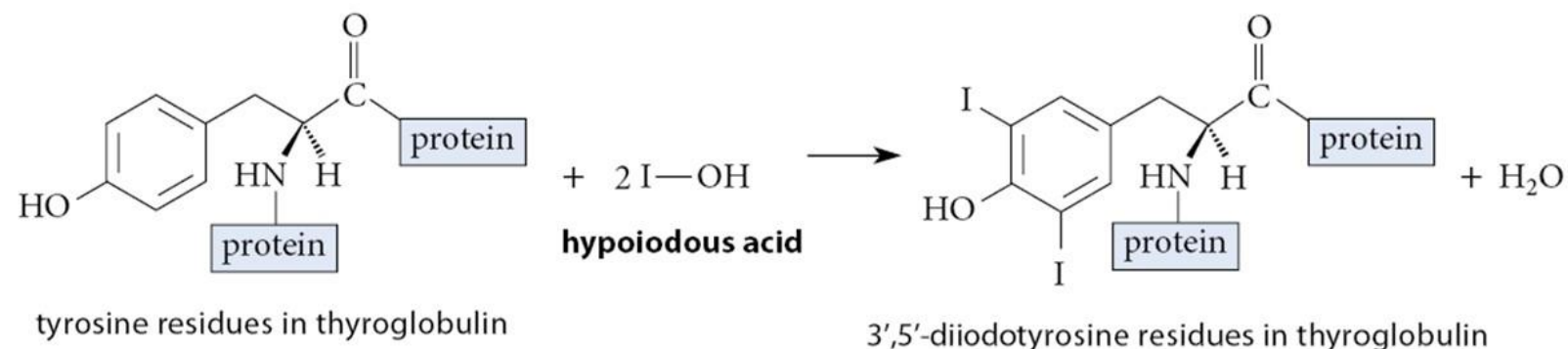
(S)-triiodothyronine (T3)
a thyroid hormone



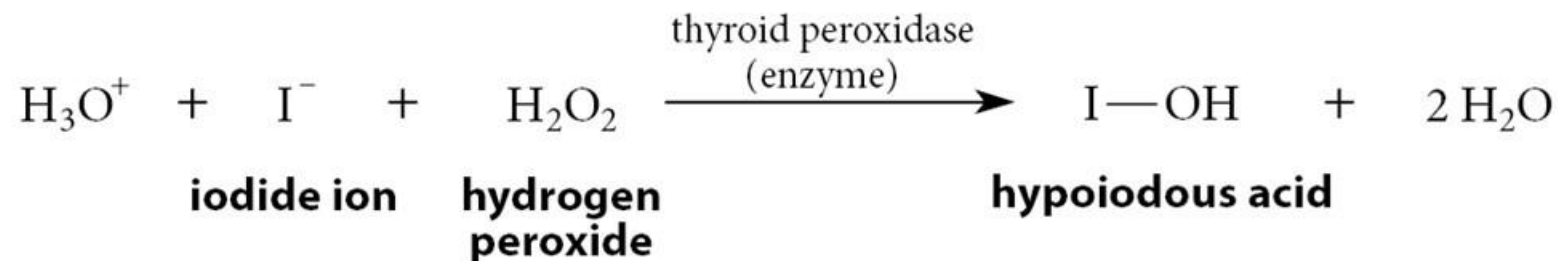
(S)-thyroxine (T4)
a thyroid hormone

Tyrosine Undergoes Iodination via Electrophilic Aromatic Substitution

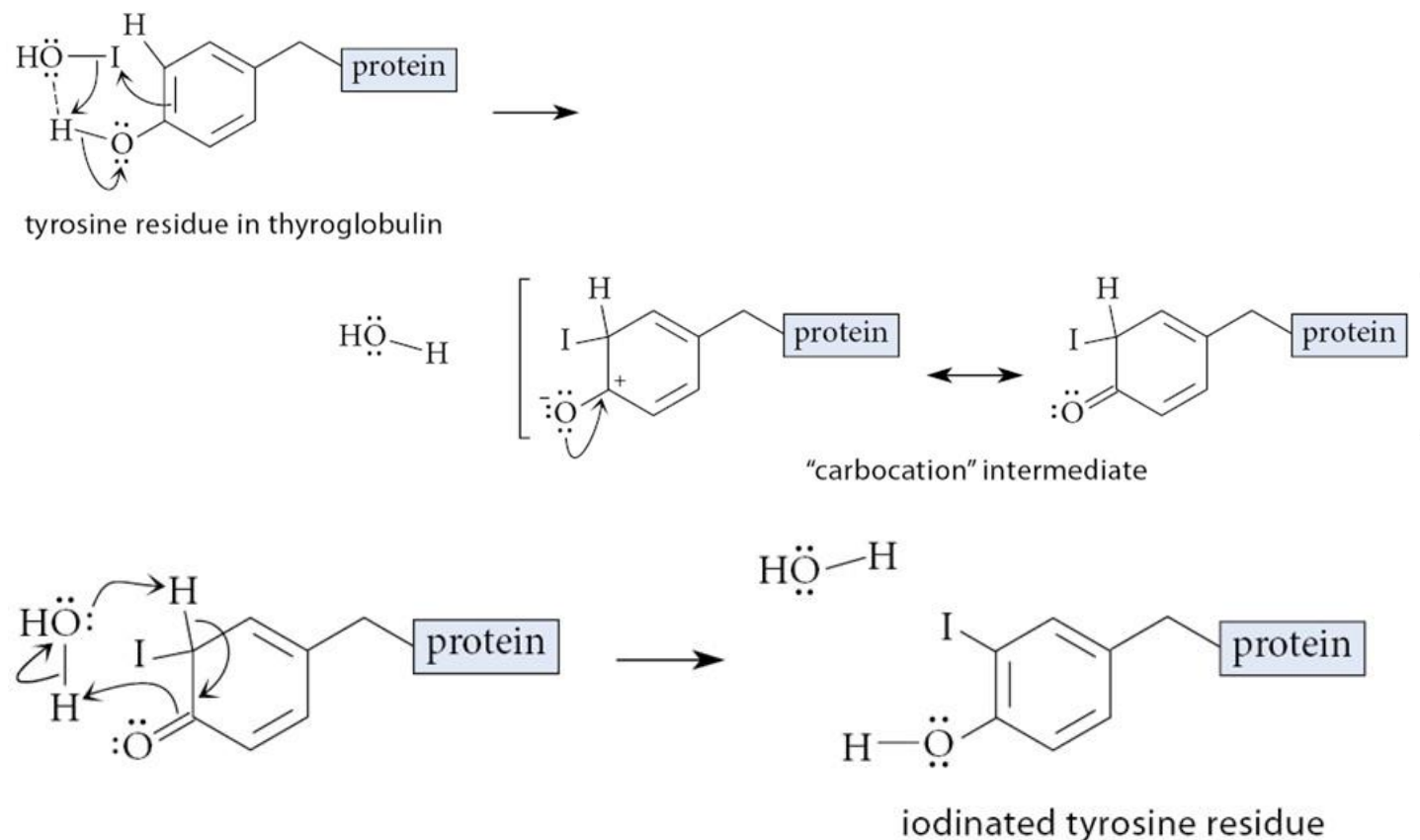
- Tyrosine residues in thyroglobulin protein undergo iodination via EAS.



- Iodine oxidized to hypoiodous acid, forming electrophile.



Mechanism of Iodination of Tyrosine in Thyroglobulin



Why halogens are deactivating?

- The only exception in the table which is like electron-donating group (o- and p-directing) but it is deactivating.
- It has lone pairs that can donate electrons to the aromatic rings. But why is it deactivating?

Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl		
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy	meta directors	deactivating substituents
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$\left. \begin{array}{l} -\text{C}(=\text{O})\text{OH} \\ -\text{C}(=\text{O})\text{NH}_2 \\ -\text{C}(=\text{O})\text{OR} \end{array} \right\}$	carboxy, carboxamido, carboalkoxy		
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Why halogens are deactivating?

- More electronegative elements pull (withdraw) electrons.

Average Pauling Electronegativities of Some Main-Group Elements

Increasing electronegativity →

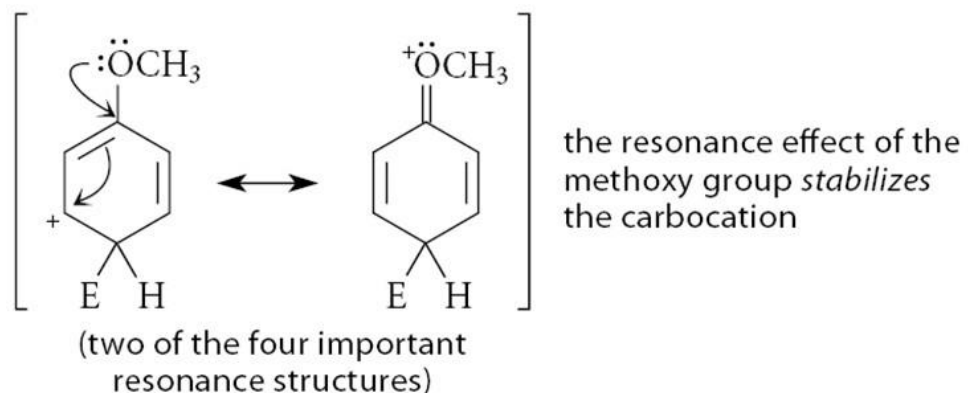
			H 2.20			
Li 0.98	Be 1.57	B 2.04	C 2.55	N 3.04	O 3.44	F 3.98
Na 0.93	Mg 1.31	Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16
K 0.82	Ca 1.00				Se 2.55	Br 2.96
Rb 0.82						I 2.66
Cs 0.79						

↑ Increasing electronegativity

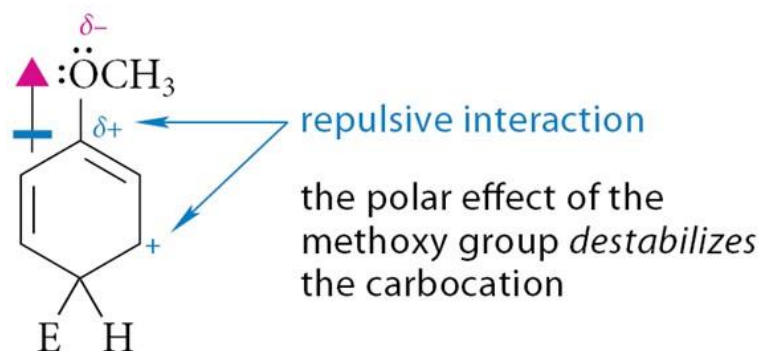
Substituent group	Name of group	Directing effect	Activating or deactivating
$-\text{NH}_2, -\text{NR}_2$	amino, alkylamino		
$-\text{OH}$	hydroxy		
$-\text{OR}$	alkoxy		
$-\text{NH}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\text{O}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\text{F}, -\text{Cl}, -\text{Br}, -\text{I}$	halogens		
$-\text{C}(=\text{O})\text{OH}, -\text{C}(=\text{O})\text{NH}_2, -\text{C}(=\text{O})\text{OR}$	carboxy, carboxamido, carboalkoxy		
$-\text{C}(=\text{O})\text{R}$	acyl	meta directors	deactivating substituents
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Explaining Activation– Deactivation

- Resonance effect:**



- Polar effect:**



Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl		
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy	meta directors	deactivating substituents
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$\left\{ \begin{array}{l} -\text{C}(=\text{O})\text{OH} \\ -\text{C}(=\text{O})\text{NH}_2 \\ -\text{C}(=\text{O})\text{OR} \end{array} \right\}$	carboxy, carboxamido, carboalkoxy		
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Which one donates electrons better?

- CH₃, NH₂, OH, or F?

Average Pauling Electronegativities of Some Main-Group Elements

Increasing electronegativity →

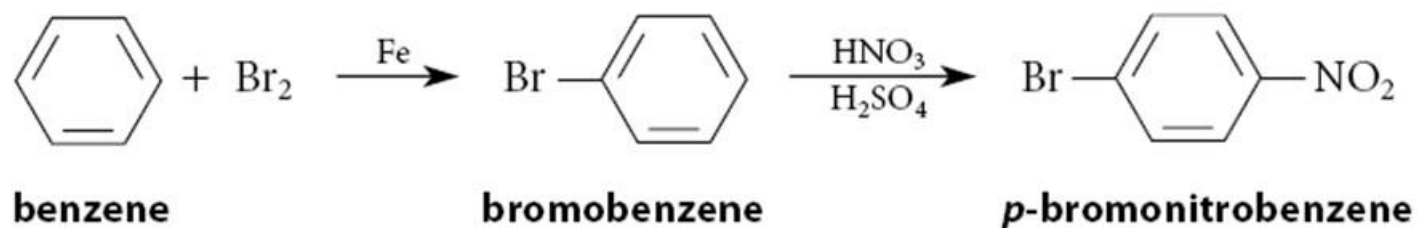
			H 2.20			
Li 0.98	Be 1.57	B 2.04	C 2.55	N 3.04	O 3.44	F 3.98
Na 0.93	Mg 1.31	Al 1.61	Si 1.90	P 2.19	S 2.58	Cl 3.16
K 0.82	Ca 1.00				Se 2.55	Br 2.96
Rb 0.82						I 2.66
Cs 0.79						

↑ Increasing electronegativity

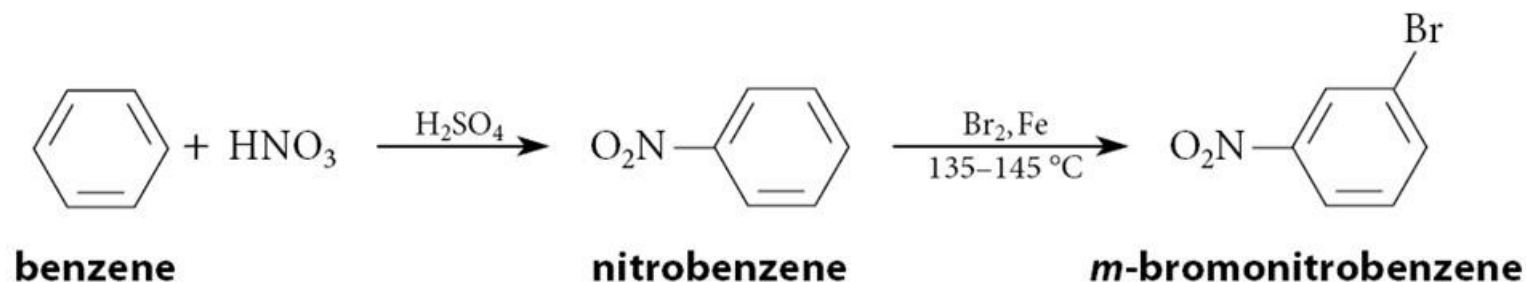
Substituent group	Name of group	Directing effect	Activating or deactivating
$-\ddot{\text{N}}\text{H}_2, -\ddot{\text{N}}\text{R}_2$	amino, alkylamino		
$-\ddot{\text{O}}\text{H}$	hydroxy		
$-\ddot{\text{O}}\text{R}$	alkoxy		
$-\ddot{\text{N}}\text{H}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl	ortho, para directors	activating substituents
$-\ddot{\text{O}}-\text{C}(=\text{O})\text{R}$	acyloxy		
	phenyl		
$-\ddot{\text{F}}:, -\ddot{\text{Cl}}:, -\ddot{\text{Br}}:, -\ddot{\text{I}}:$	halogens		
$-\text{C}(=\text{O})\text{OH}, -\text{C}(=\text{O})\text{NH}_2, -\text{C}(=\text{O})\text{OR}$	carboxy, carboxamido, carboalkoxy		
$-\text{C}(=\text{O})\text{R}$	acyl	meta directors	deactivating substituents
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Study Problem 16.2

- Outline a synthesis of *p*-bromonitrobenzene from benzene:
 - Correct sequence

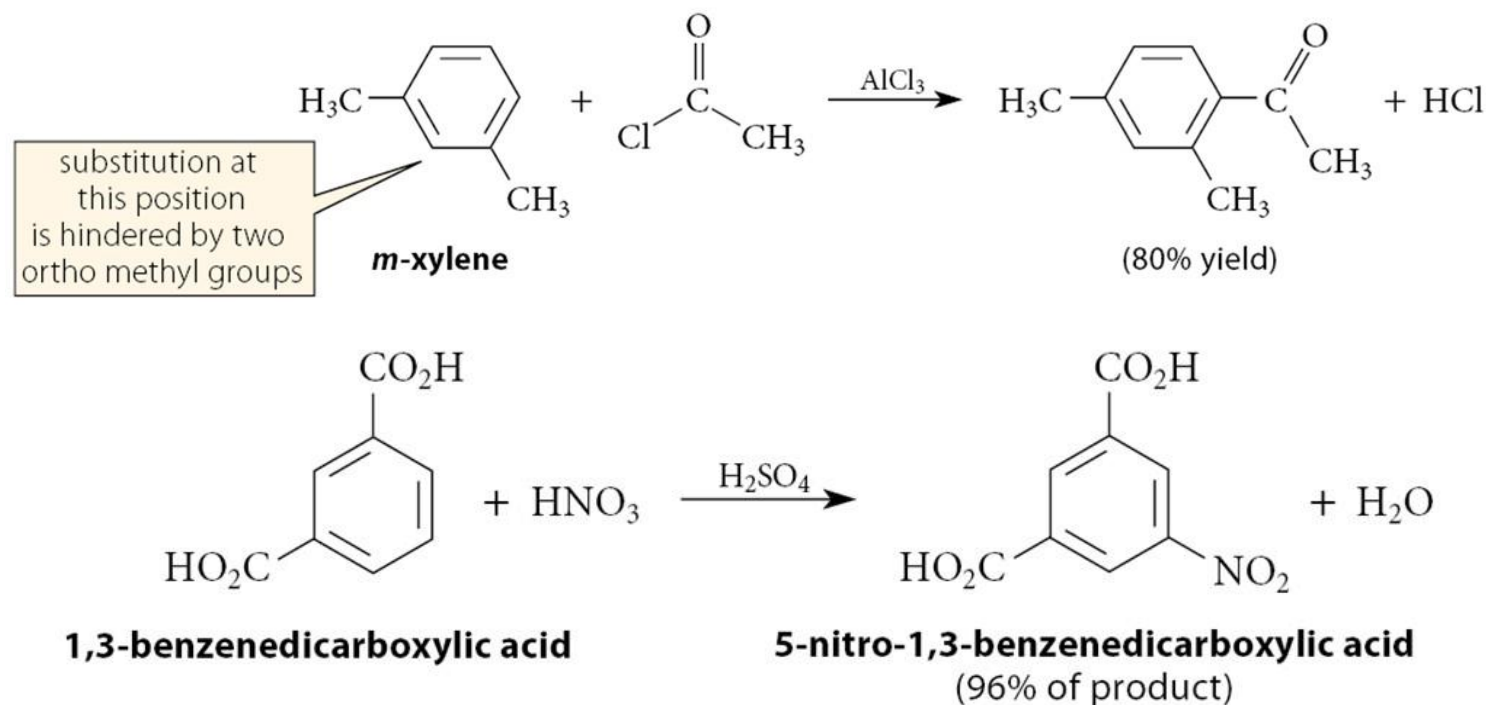


- Incorrect sequence



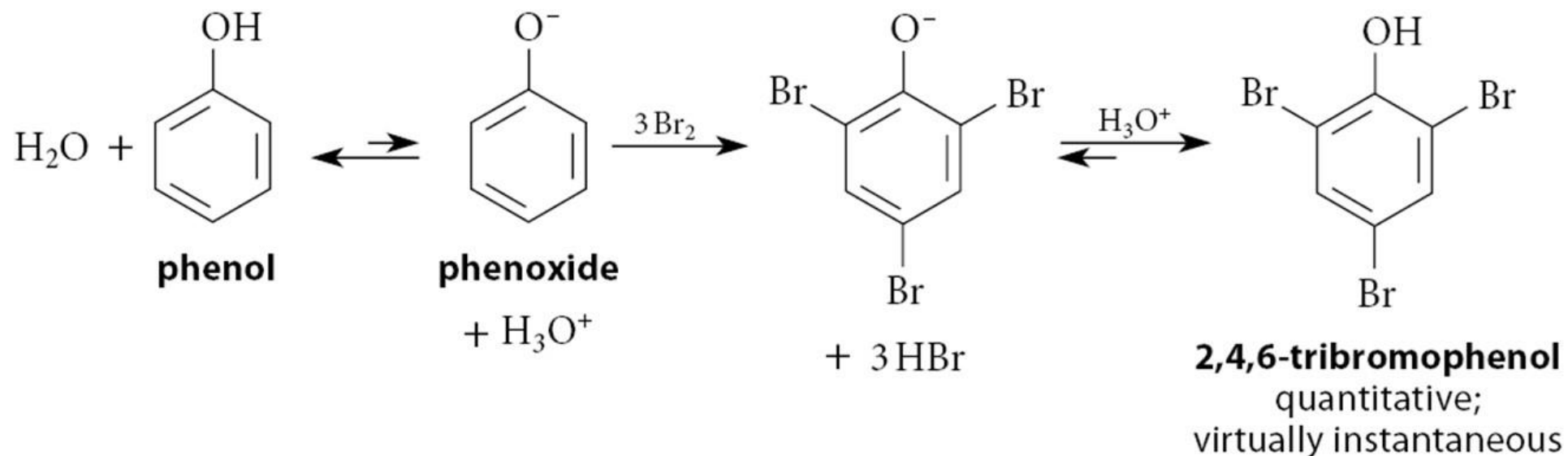
Effect of Two or More Substituents

- With two or more substituents, the activating and directing effects are roughly the sum of the effects of the individual substituents.



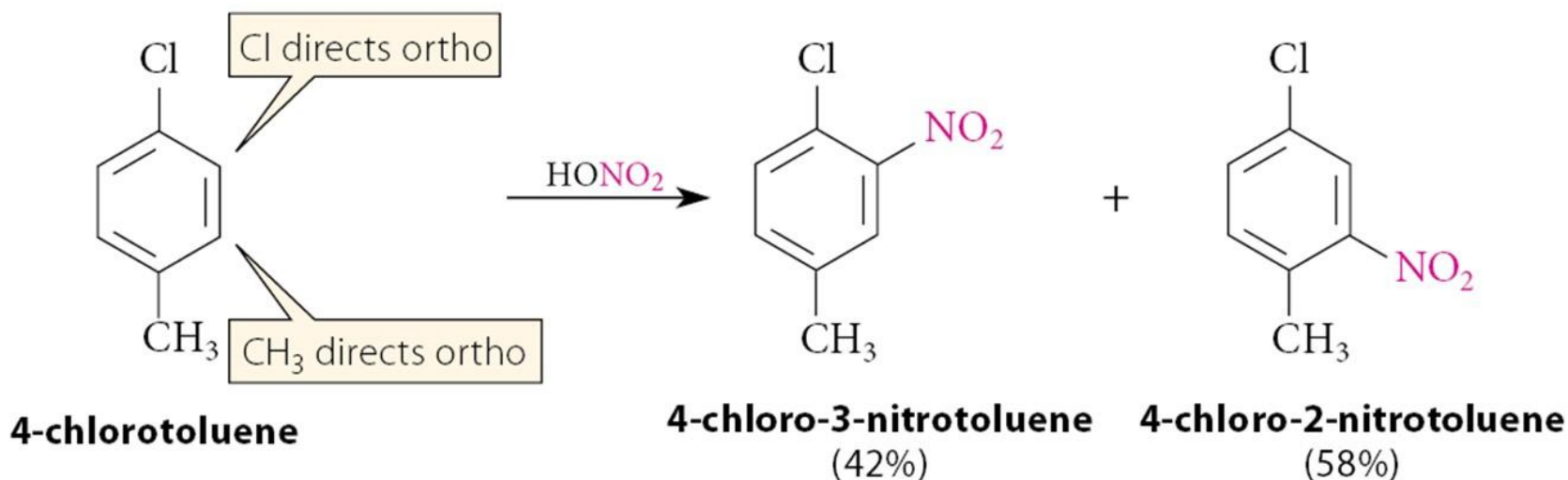
The -O^- group is a Powerful Activating Group

- Phenol oxidizes to phenoxide when treated with bromine in water, allowing it to undergo *three* bromination reactions.



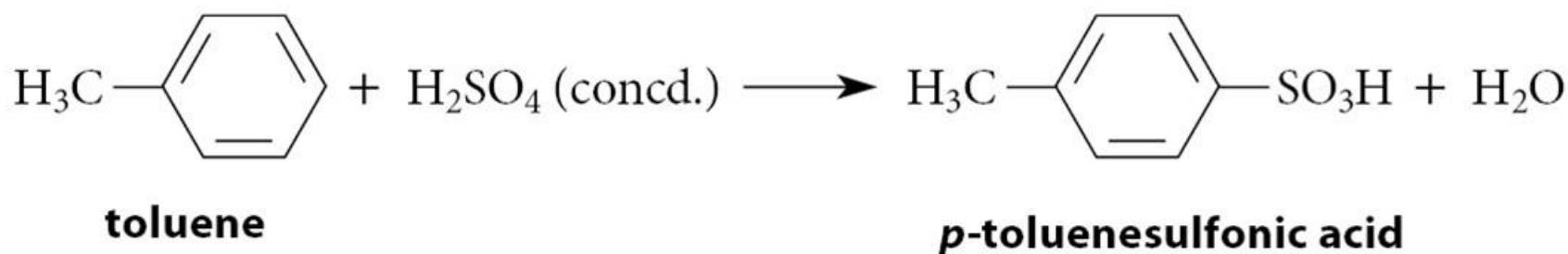
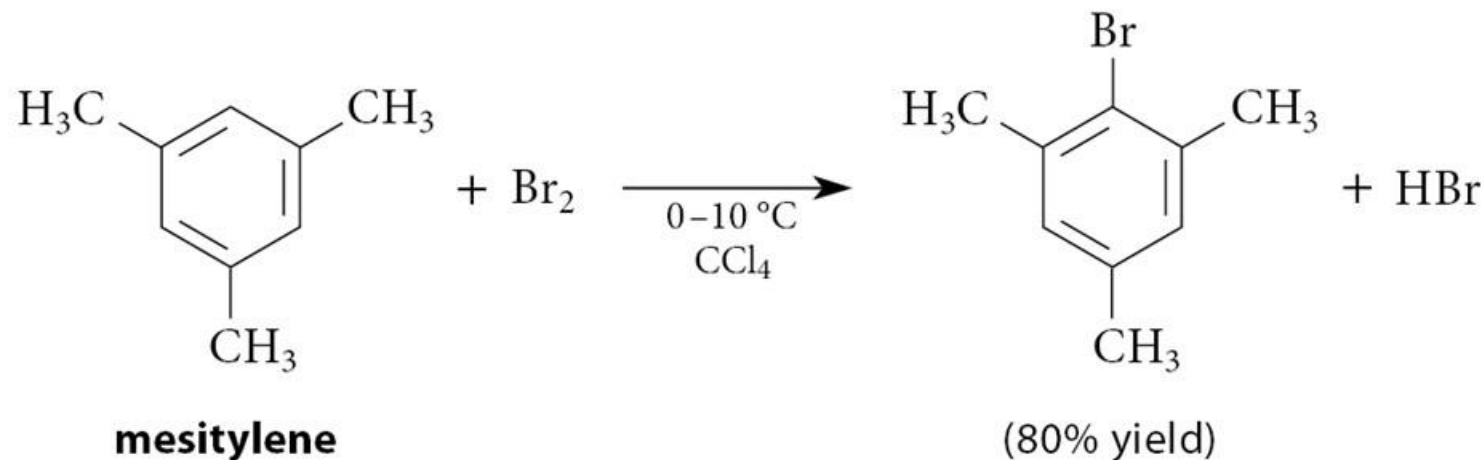
Use of Electrophilic Aromatic Substitution in Organic Synthesis

- Mixtures are typical in some cases:



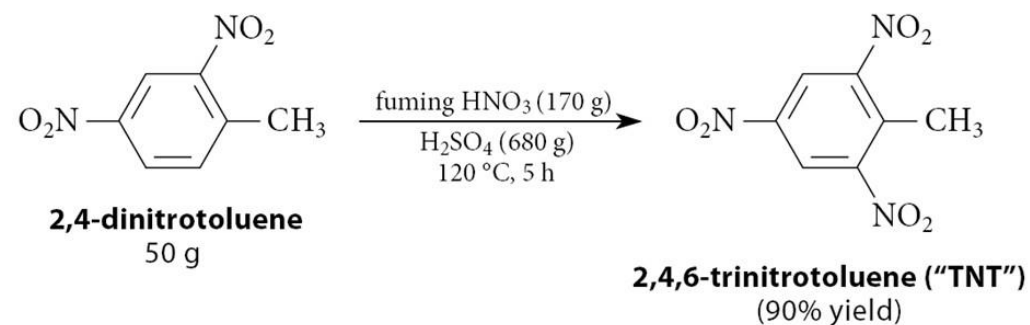
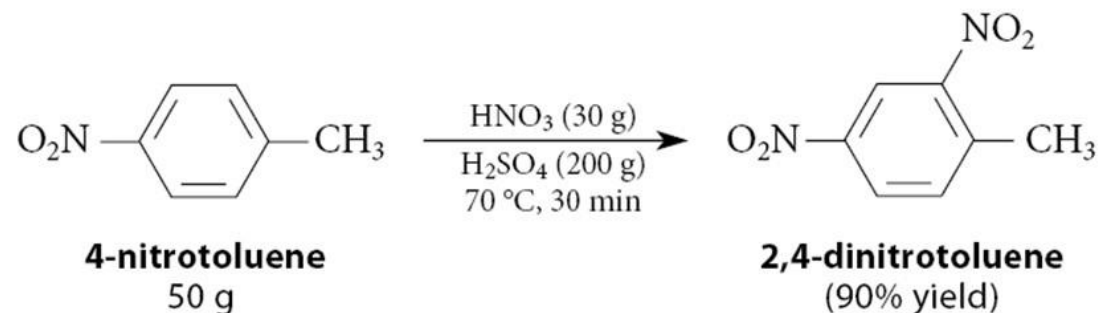
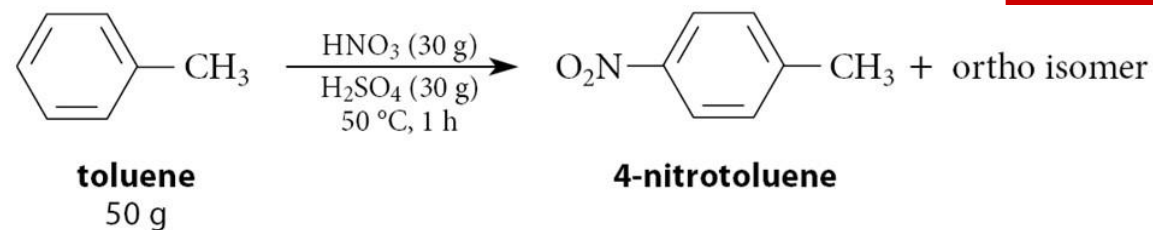
Substituent group	Name of group	Directing effect	Activating or deactivating
$-\text{NH}_2, -\text{NR}_2$	amino, alkylamino	ortho, para directors	activating substituents
$-\text{OH}$	hydroxy		
$-\text{OR}$	alkoxy		
$-\text{NH}-\text{C}(=\text{O})\text{R}$	acylamino		
$-\text{R}$	alkyl		
$-\text{O}-\text{C}(=\text{O})\text{R}$	acyloxy	meta directors	deactivating substituents
$-\text{C}_6\text{H}_5$	phenyl		
$-\text{F}, -\text{Cl}, -\text{Br}, -\text{I}$	halogens		
$-\text{C}(=\text{O})\text{OH}, -\text{C}(=\text{O})\text{NH}_2, -\text{C}(=\text{O})\text{OR}$	carboxy, carboxamido, carboalkoxy		
$-\text{C}(=\text{O})\text{R}$	acyl		
$-\text{SO}_3\text{H}$	sulfonic acid		
$-\text{C}\equiv\text{N}$	cyano		
$-\text{NO}_2$	nitro		

Activating Groups Allow Some Electrophilic Aromatic Substitution Reactions to Occur Without Catalyst



Adding Deactivating Groups

- Deactivating groups produce less reactive products.
- Easier to introduce one group at a time.



Deactivating Groups Can Be Too Deactivating

